

Exercise: Use the Drawing Tools and the Library Toolbar

Purpose

- Learn how to use standard descriptors.
- Learn how to connect them to IGSS objects.
- Learn how to use the **Library** toolbar.

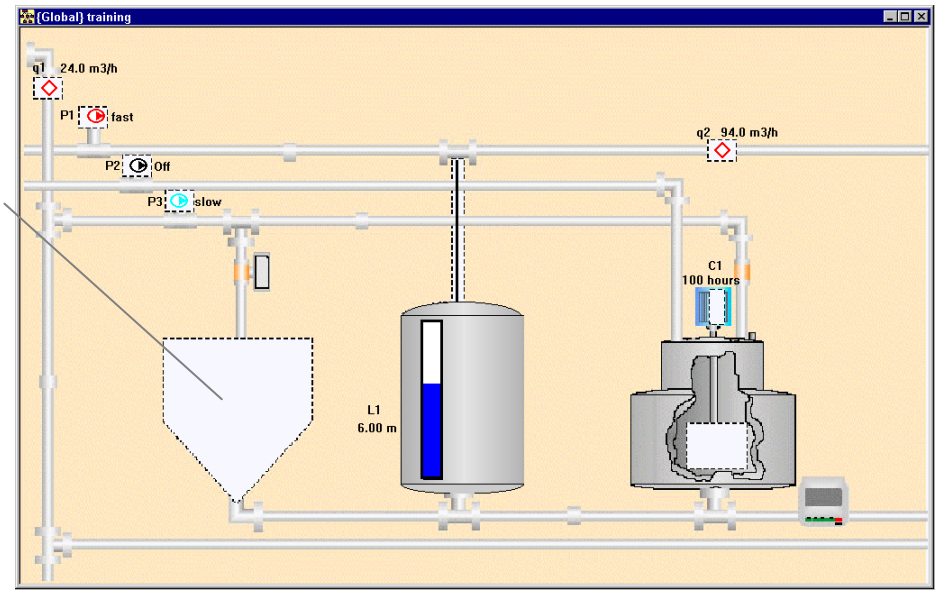
Duration

Max. 30 minutes.

Task 1: Connect an analog object to a polygon

We will create a new level gauge (**L2**), which we base on the **LEVEL** template, and connect a polygon (which is one of the standard descriptors) to it. The filling of the polygon will reflect the current value of the water level in the tank.

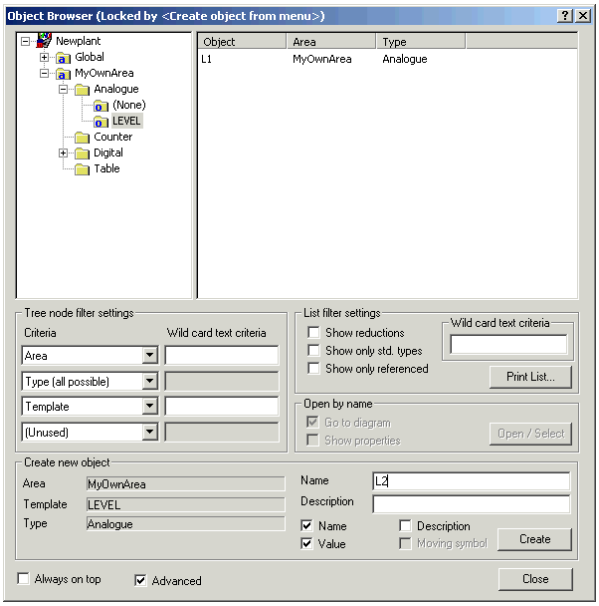
Position **L2** here



Step	Action
1.	*TIP:* On the **Objects** menu, select **Standard Descriptors** → **Polygon…** which brings up the **Object Browser** dialog box.
2.	Click the **+** sign beside **Workshop**, the **Analog** and finally under **Template**, select **LEVEL**.

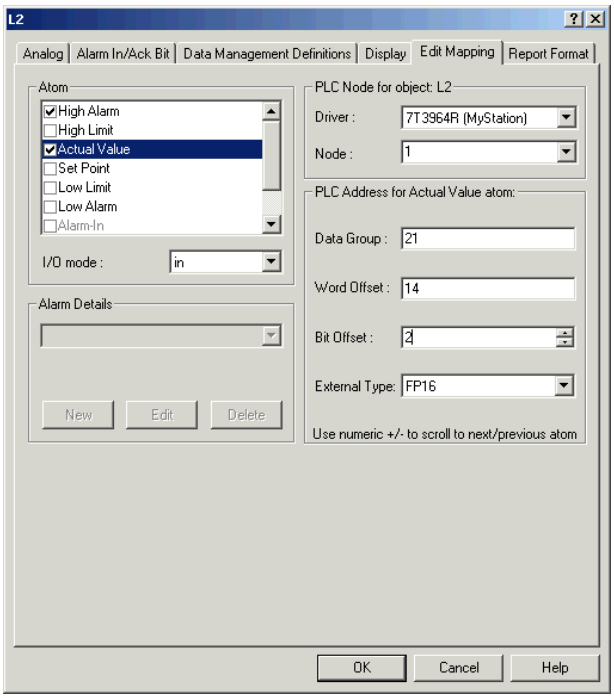


3. In the **Name** field in the lower right corner, type `L2`.



4. Click **Create**.

5. Click on the **Edit Mapping** tab and then click on the **Actual Value** atom. Look to the right under the text box **PLC Address for Actual Value atom** and type in a unique PLC address by for instance typing **2** in **Bit Offset:**.



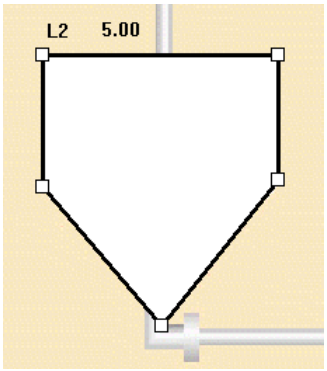
6. Click **OK.**
(The polygon descriptor now appears in the upper left corner of the diagram as a triangle).



7. On the **Format** menu, select **Snap to Grid** to make positioning of graphics on the diagram easier.

8. Now drag the polygon "triangle" into the cutout for **L2**. Double-click on one of the handles - which appear as squares - to add another handle. Do this one more time to get five handles altogether.

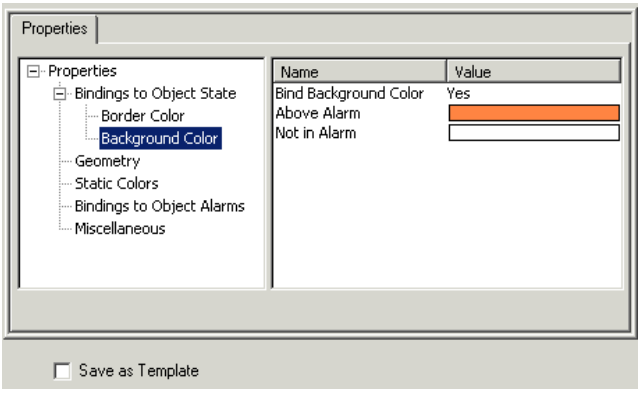
Move the five handles so that they fill out the entire cutout.

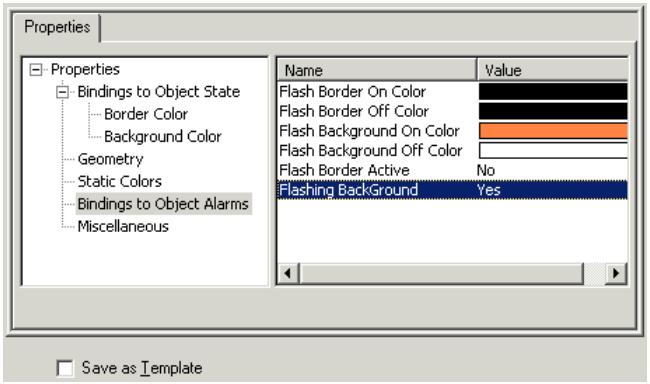


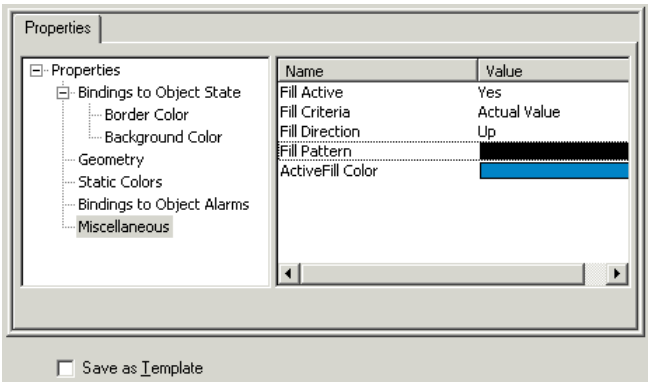
9. Double-click on one of the polygon lines to bring up the object dialog box. Then click on the tab **Attributes of Polygon** and choose the properties and settings shown in the following screen shots.
(We want to represent the tank filling up).

We use only three properties here: **Background Color**, **Bindings to Object Alarms** and **Miscellaneous**.

TIP: Click on the column edge between **Name** and **Value** and move it slightly to the right to be able to see the complete text under **Name**.



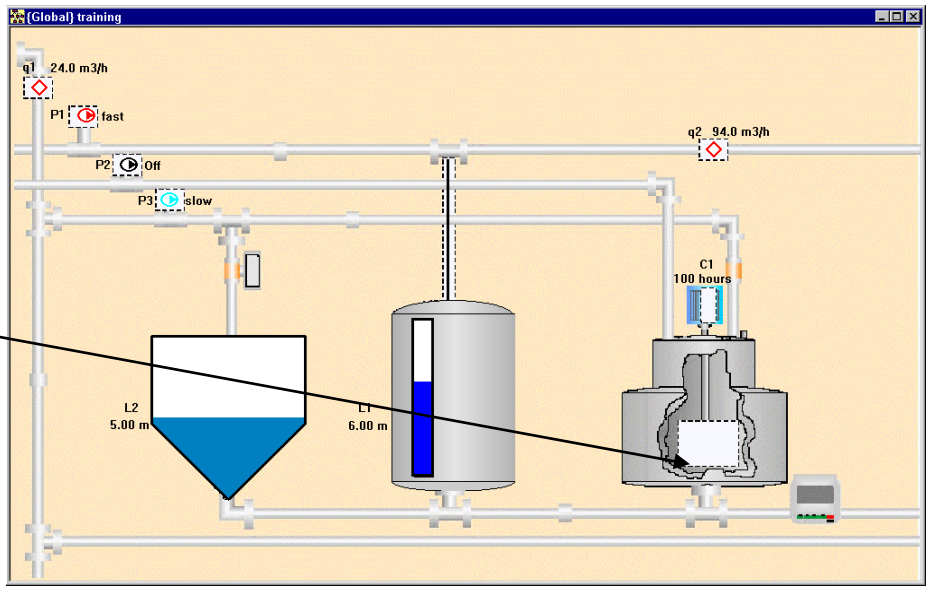




Task 2: Use the Library toolbar

We will copy the tank we inserted earlier on the diagram to the **Library Toolbar**. After that we will first add a new graphic (**Prop-1.gif**) to the **Library Toolbar** and then drag it onto the diagram.

Position **Prop_1.gif** here



Step	Action
1.	On the **View** menu, select **Library Toolbar** (the toolbar appears on top of the diagram).



2. Drag the tank graphic to the toolbar by dragging it to the **Library Toolbar.** Drop it inside the toolbar.

3. Right-click on the graphic and choose **Edit Tool Name…** Give the tank the name, **Tank2**.

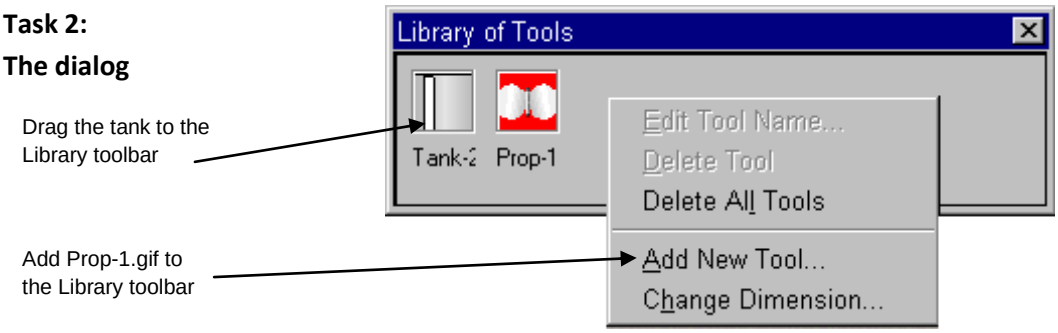
4. *TIP:* Add **Prop-1.gif** as a new tool by right-clicking inside the **Library Toolbar** and selecting **Add New Tool**.

5. Give the picture the name, **Prop-1**.

6. Drag **Prop-1** onto the diagram and place it as shown in the figure above.

Task 2:

The dialog





Task 3: Create a digitial template for a 2 state motor

We will create a motor object called and represent this process component on the mimic with a gif file instead of using a standard symbol. The motor object is created from a new template called **MOTOR**, which we first have to create.

Step	Action
1.	Create a new ***digital*** template called **MOTOR** with the following properties:

Create Template	**Name:** **Description:**	**MOTOR** **2-state motor**
Tab Name	***Property***	***Setting***
Bit Map I/O	**To PC:** **State:**	Set bit **0** as state bit.
	From PC: **Command:**	Set bit **0** as command bit.
Alarm In/Ack Bit	**Alarm bits**	Under **To PC**, set bit **0** as alarm indication bit (**+**) and set bit **1** as alarm acknowledgement bit (*****). Under **From PC**, set bit **0** as alarm acknowledgement bit (*****) to PC.
States and Commands	**State**	Off, on
	Command	Stop, start
Data Management Definitions	**Scan interval**	5000 msec.
	Logging	All changes
	Base interval	10 min.
	Data reduction	Change
	Transfer to History	Actual Total Value



Command/ State Config	**States Commands**	Set the appropriate command to be available for each state.
Edit Mapping 	**PLC addresses for:**	**Command:** DG = 25, WO = 10, BO = 0 (I/O mode must be "Out"). **State:** DG = 26, WO = 10, BO = 0 (I/O mode must be "In") **Alarm-in:** DG = 27, WO = 10, BO = 0 (I/O mode must be "In") Alarm-out: DG = 28, WO = 10, BO = 0 (I/O mode must be "Out")
	Alarm details	Select the **Alarm-in** atom and attach the new alarm text “Motor defective”, priority = 6, alarm color = red and blinking enabled, operator instructions “Call maintenance team.”

Step	Action
2.	Click OK to complete the Edit Mapping settings and then click OK again to save and exit the **MOTOR** template.

Task 4:Create the motor, M1 and connect itm to a gif file

Now we must create a motor object called **M1**, which is based on the **MOTOR** template. Because we don't want to use a standard symbol from the IGSS standard symbol library to represent the motor, we first create it ***without*** a symbol representation.

1. *TIP:* Create the **M1** object by first selecting from the **Objects** menu the entry called **New Unreferenced**, which is found vertically down in the menu structure.

2. In the **Object Browser**, fold out the **Workshop** area in the tree view at the left (click on the + sign) and then fold out the **Digital** folder (click on the + sign).

3. Select the template name **MOTOR** and in the lower right portion of the **Browser** in the **Name** field key in the name **M1**.



4.	Click the button **Create** and the properties tabs for the **M1** motor come up.
5.	Go to the Edit Mapping tab, select the **Atom** called **Command** and in the **PLC Addresses** group on the right, key in `10` under **Bit Offset**.
6.	Click **OK** to complete. NOTE: There is no graphical representation on the mimic for **M1** yet.

Task 5:Connect the gif graphic to the M1 object.

Finally, we connect the **Prop_1.gif** to the **M1** motor to give it a graphical representation on the mimic.

1.	*TIP:* Right click on the **Prop_1.gif** graphic and select **Connect…**.
2.	In the **Object Browser** find the object **M1** in the object list on the right and select it.
3.	In the **Open by name** area select **Go to diagram** and **Show properties** and then click the button **Open / Select**.
4.	The **M1** object properties tabs come up; choose the **Attributes of GIF** tab.
6.	Unfold the **Properties** branch on the left by clicking on the + sign and then unfold the **Bindings to Object State** in the same way..
7.	Select the **Play** branch and put in the following settings by double clicking on the entry under **Value**: **Bind Play Yes** **Off No** **On Yes**
8.	Select the **Miscellaneous** branch and beside the **Play Non Stop** entry, double click the value to change it to **Yes**.
9.	Click **OK** to save and exit.

YOU HAVE SUCCESSFULLY COMPLETED THE EXERCISE !!